

EXPLOITING PRIOR-PHASE EFFORT DATA TO ESTIMATE THE EFFORT FOR THE SUBSEQUENT PHASES: A FURTHER ASSESSMENT

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Outline

- Motivation
 - Use of prior-phase effort data to estimate the effort of subsequent phases
 - Previous similar studies
- The goal of our study
- Empirical study
 - Design
 - Results and Discussion
 - Threats to validity
- Future work

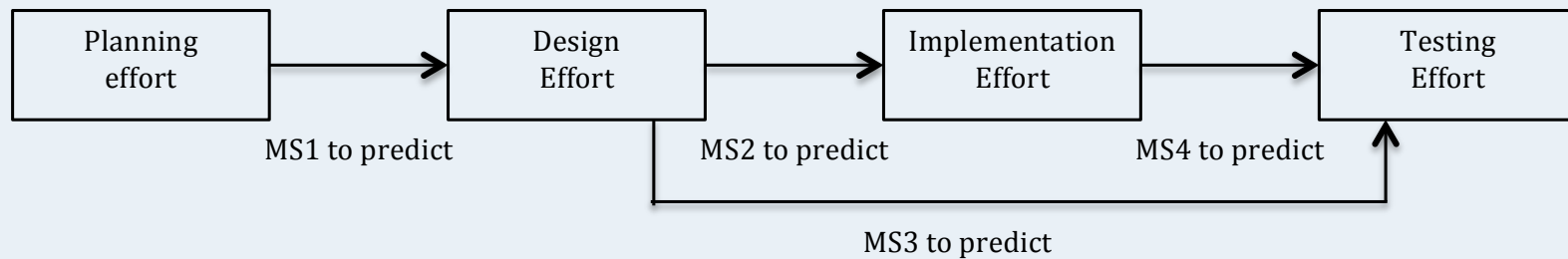
Motivation

- Effort estimation is crucial to allow projects to be delivered on time and within budget
- FSM methods (e.g., FPA) are widely applied for sizing software systems
 - Size is then exploit to estimate the effort
 - $\text{Effort} = f(\text{FPA size})$
- A few works have investigated the possibility of using as predictor the effort of activities already completed
 - $\text{Effort}_{\text{phase}_X} = f(\text{Effort}_{\text{previous_phases_of_X}})$

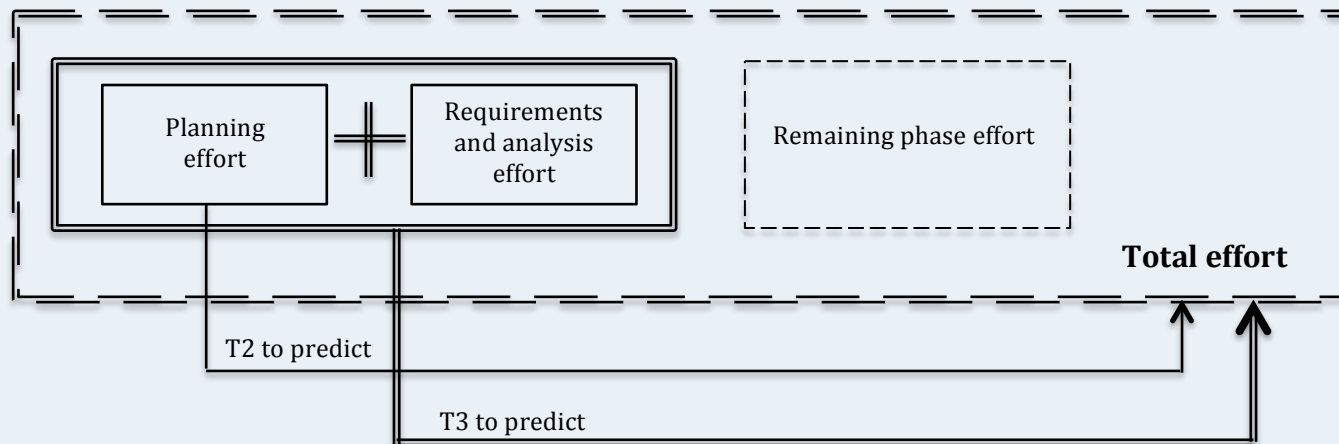
Motivation (2)

- Effort estimation can benefit from the information that become available as the project progresses
- A continuous estimation is required to assess whether the initial estimation has been fairly good or it has to be adjusted according to the actual status of the project
- Project managers can benefit from updated estimations not only of the total project effort but also of the effort devoted to each phase

Previous studies



(a) MacDonell and Shepperd analysis [27]

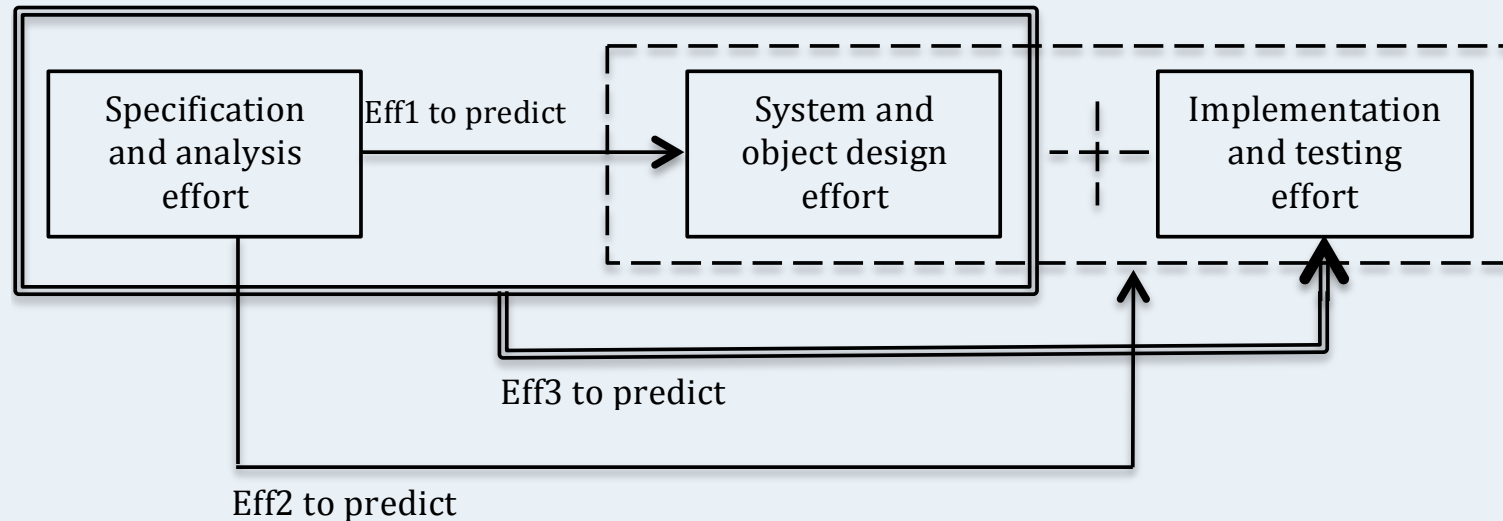


(b) Tsunoda et al. Analysis [41]

[27] MacDonell and Shepperd, "Using prior-phase effort records for re-estimation during software projects", ESEM 2003

[41] Tsunoda et al. "Revisiting software development effort estimation based on early phase development activities." MSR 2013

Our goal



(c) our analysis

- Assess and compare Eff1, Eff2, Eff3 against Function Points based estimation models (FP1, FP2, FP3)
- Assess a combination of prior phase efforts and Function Points to build estimation models (EffFP1, EffFP2, EffFP3)

Research questions

- **RQ1:** Can prior-phase efforts provide more accurate estimates of the subsequent phase(s) effort than the corresponding models based on the use of Function Points alone?
- **RQ2:** Can the combination of Function Points and prior- phase efforts provide more accurate estimates of the subsequent phase(s) efforts than the corresponding models based on the use of prior-phase efforts alone?

Data set

- Data for our study provided by an Italian software company
 - Specialized in the design, development, and management of solutions for Web portals, enterprise intranet/extranet applications (such as Content Management Systems, e-commerce, work-flow managers, etc.), and GIS.
 - certified ISO 9001:2000, and also a certified partner of Microsoft, Oracle, and ESRI
- The company provided us a set of **25 applications**
 - e.g., e-government, e-banking, Web portals, and Intranet applications
 - developed by exploiting a wide range of technologies, such as J2EE, ASP.NET, etc.

Variables

- Data obtained in a long term investigation aimed at verifying the effectiveness of FSM methods in estimating the whole development effort

DESCRIPTIVE STATISTICS

Var	Obs	Min	Max	Mean	Median	Std. Dev.
EFF_A	25	208	1496	853.2	864	332.04
EFF_D	25	264	1528	905	960	352.38
EFF_I	25	310	1513	817.8	738	319.43
$EFF_{D\&I}$	25	574	3041	1724	1748	664.24
FP	25	89	915	366.76	303.94	208.65

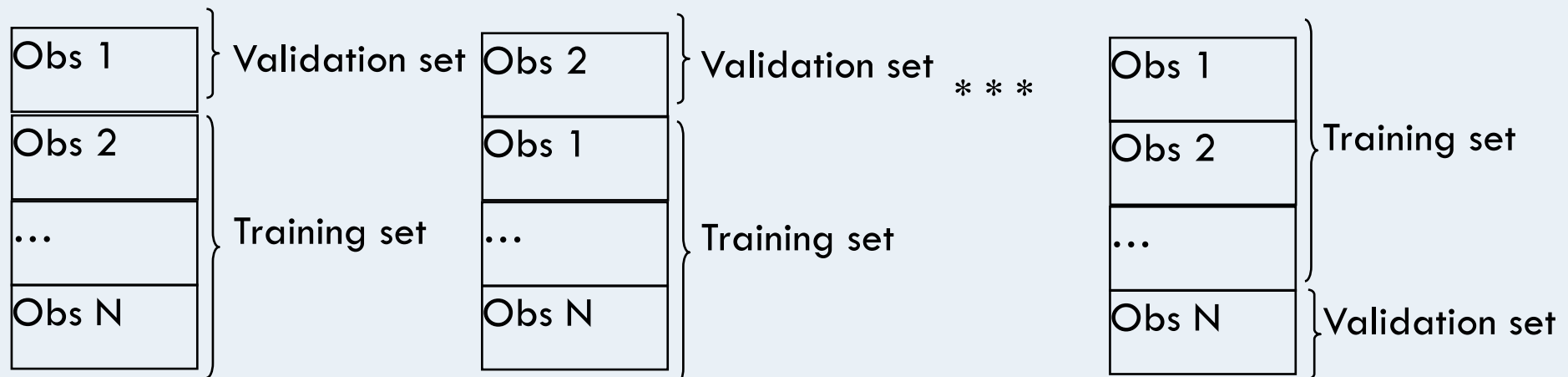
Estimation technique

- **Simple Linear Regression (SLR):** 1 dependent var. and 1 independent var., e.g., $EFF_D = a * EFF_A + b$ or $EFF_D = a * FP + b$
- **StepWise linear Regression (SWR):** 1 dependent var. and more independent var. computed in stages:
 - at each stage, it is added the independent var. with the highest association to the dependent variable, taking into account all the variables currently in the model.
 - to find the set of independent variables (predictors) that best explains the variation in the dependent variable (response)
 - $EFF_D = a_1 * EFF_A + a_2 * FP + b$
 - The automatic procedure of R environment

Validation method

- **Leave-1-out cross validation**

- $N=25$ different subsets of training (TS) and validation (VS) sets, where each validation set had one observation.
- TS used to build the estimation model with linear regression and VS used to assess the obtained estimates



Evaluation criteria

- **Mean and Median of Absolute Residuals (MAR and MdAR)**
Absolute Residual = (Actual effort— Predicted effort)
- **Wilcoxon Test:** to verify the statistical significance of results
 - For each test we decided to accept a probability of 5% of committing a Type-I-Error
- **Effect size:** to quantify the magnitude of the difference
 - We applied Cliff'd
 - Effect size classified as follows: negligible, small, medium, and large

Results: estimation models

RQ	Dependent variable	Independent variables	Model	Variable(s)/ Intercept	Value	Std. Err	t-value	p-value	R^2	Std. Err	F	Sign. F
RQ1	EFF _D	EFF _A	Eff1	EFF _A Intercept	1.029 27.948	0.054 49.489	18.98 0.56	<0.001 0.58	0.94	88.2	360	<0.001
	EFF _D	FP	FP1	FP Intercept	1.309 425.505	0.222 93.369	5.89 4.56	<0.001 <0.001	0.601	227	34.7	<0.001
	EFF _{D&I}	EFF _A	Eff2	EFF _A Intercept	1.927 79.452	0.112 102.124	17.230 0.780	<0.001 0.440	0.928	182	297	<0.001
	EFF _{D&I}	FP	FP2	FP Intercept	2.597 771.244	0.384 161.216	6.760 4.780	<0.001 <0.001	0.665	392	45.800	<0.001
	EFF _I	EFF _A , EFF _D	Eff3	EFF _D Intercept	0.865 34.235	0.057 54.822	15.30 0.62	<0.01 0.58	0.91	97.6	234	<0.001
	EFF _I	FP	FP3	FP Intercept	1.287 345.634	0.173 72.563	7.453 4.76	<0.001 <0.001	0.707	177	55.5	<0.001
RQ2	EFF _D	EFF _A , FP	EffFP1(=Eff1)	EFF _A Intercept	1.029 27.948	0.054 49.489	18.98 0.56	<0.001 0.58	0.94	88.2	360	<0.001
	EFF _{D&I}	EFF _A , FP	EffFP2	EFF _A FP Intercept	1.627 0.629 105.113	0.155 0.246 92.223	10.53 2.560 1.140	<0.001 0.018 0.267	0.945	163	188	<0.001
	EFF _I	EFF _A , EFF _D , FP	EffFP3	EFF _D FP Intercept	0.687 0.387 34.235	0.077 0.131 54.822	8.87 2.96 0.62	<0.001 0.007 0.58	0.936	84.4	161	<0.001

- Linear regression assumptions preliminarily verified
- R^2 and corresponding p-value (Sign. F) for the goodness of fit; t-statistics to assess the significance of predictors

Results: estimation models (2)

RQ	Dependent variable	Independent variables	Model	Variable(s)/ Intercept	Value	Std. Err	t-value	p-value	R^2	Std. Err	F	Sign. F
RQ1	EFF _D	EFF _A	Eff1	EFF _A	1.029	0.054	18.98	<0.001	0.94	88.2	360	<0.001
				Intercept	27.948	49.489	0.56	0.58				
	EFF _D	FP	FP1	FP	1.309	0.222	5.89	<0.001	0.601	227	34.7	<0.001
				Intercept	425.505	93.369	4.56	<0.001				
	EFF _{D&I}	EFF _A	Eff2	EFF _A	1.927	0.112	17.230	<0.001	0.928	182	297	<0.001
				Intercept	79.452	102.124	0.780	0.440				
RQ2	EFF _{D&I}	FP	FP2	FP	2.597	0.384	6.760	<0.001	0.665	392	45.800	<0.001
				Intercept	771.244	161.216	4.780	<0.001				
	EFF _I	EFF _A , EFF _D	Eff3	EFF _D	0.865	0.057	15.30	<0.01	0.91	97.6	234	<0.001
				Intercept	34.235	54.822	0.62	0.58				
	EFF _I	FP	FP3	FP	1.287	0.173	7.453	<0.001	0.707	177	55.5	<0.001
				Intercept	345.634	72.563	4.76	<0.001				
RQ2	EFF _D	EFF _A , FP	EffFP1(=Eff1)	EFF _A	1.029	0.054	18.98	<0.001	0.94	88.2	360	<0.001
				Intercept	27.948	49.489	0.56	0.58				
	EFF _{D&I}	EFF _A , FP	EffFP2	EFF _A	1.627	0.155	10.53	<0.001	0.945	163	188	<0.001
				FP	0.629	0.246	2.560	0.018				
				Intercept	105.113	92.223	1.140	0.267	0.936	84.4	161	<0.001
	EFF _I	EFF _A , EFF _D , FP	EffFP3	EFF _D	0.687	0.077	8.87	<0.001				
				FP	0.387	0.131	2.96	0.007				
				Intercept	34.235	54.822	0.62	0.58				

- high R^2 and F values and Sign.F <0.05: the prediction is indeed possible with a high degree of confidence
- predictors important and significant (t-value >1.5 and p-value<0.05)

Results: estimation models (3)

RQ	Dependent variable	Independent variables	Model	Variable(s)/ Intercept	Value	Std. Err	t-value	p-value	R^2	Std. Err	F	Sign. F
RQ1	EFF _D	EFF _A	Eff1	EFF _A Intercept	1.029 27.948	0.054 49.489	18.98 0.56	<0.001 0.58	0.94	88.2	360	<0.001
	EFF _D	FP	FP1	FP Intercept	1.309 425.505	0.222 93.369	5.89 4.56	<0.001 <0.001	0.601	227	34.7	<0.001
	EFF _{D&I}	EFF _A	Eff2	EFF _A Intercept	1.927 79.452	0.112 102.124	17.230 0.780	<0.001 0.440	0.928	182	297	<0.001
	EFF _{D&I}	FP	FP2	FP Intercept	2.597 771.244	0.384 161.216	6.760 4.780	<0.001 <0.001	0.665	392	45.800	<0.001
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	EFF _I	FP	FP3	FP Intercept	1.287 345.634	0.173 72.563	7.453 4.76	<0.001 <0.001	0.707	177	55.5	<0.001
RQ2	EFF _D	EFF _A , FP	EffFP1(=Eff1)	EFF _A Intercept	1.029 27.948	0.054 49.489	18.98 0.56	<0.001 0.58	0.94	88.2	360	<0.001
	EFF _{D&I}	EFF _A , FP	EffFP2	EFF _A FP Intercept	1.627 0.629 105.113	0.155 0.246 92.223	10.53 2.560 1.140	<0.001 0.018 0.267	0.945	163	188	<0.001
	EFF _I	EFF _A , EFF _D , FP	EffFP3	EFF _D FP Intercept	0.687 0.387 34.235	0.077 0.131 54.822	8.87 2.96 0.62	<0.001 0.007 0.58	0.936	84.4	161	<0.001

- Lower R^2 and F values but Sign.F <0.05: the prediction is indeed possible with a high degree of confidence
- predictors important and significant (t-value >1.5 and p-value<0.05)

Results: estimation models (4)

RQ	Dependent variable	Independent variables	Model	Variable(s)/ Intercept	Value	Std. Err	t-value	p-value	R^2	Std. Err	F	Sign. F
RQ1	EFF _D	EFF _A	Eff1	EFF _A Intercept	1.029 27.948	0.054 49.489	18.98 0.56	<0.001 0.58	0.94	88.2	360	<0.001
	EFF _D	FP	FP1	FP Intercept	1.309 425.505	0.222 93.369	5.89 4.56	<0.001 <0.001	0.601	227	34.7	<0.001
	EFF _{D&I}	EFF _A	Eff2	EFF _A Intercept	1.927 79.452	0.112 102.124	17.230 0.780	<0.001 0.440	0.928	182	297	<0.001
	EFF _{D&I}	FP	FP2	FP Intercept	2.597 771.244	0.384 161.216	6.760 4.780	<0.001 <0.001	0.665	392	45.800	<0.001
	EFF _I	EFF _A , EFF _D	Eff3	EFF _D Intercept	0.865 34.235	0.057 54.822	15.30 0.62	<0.01 0.58	0.91	97.6	234	<0.001
	EFF _I	FP	FP3	FP Intercept	1.287 345.634	0.173 72.563	7.453 4.76	<0.001 <0.001	0.707	177	55.5	<0.001
RQ2	EFF _D	EFF _A , FP	EffFP1(=Eff1)	EFF _A Intercept	1.029 27.948	0.054 49.489	18.98 0.56	<0.001 0.58	0.94	88.2	360	<0.001
	EFF _{D&I}	EFF _A , FP	EffFP2	EFF _A FP Intercept	1.627 0.629 105.113	0.155 0.246 92.223	10.53 2.560 1.140	<0.001 0.018 0.267	0.945	163	188	<0.001
	EFF _I	EFF _A , EFF _D , FP	EffFP3	EFF _D FP Intercept	0.687 0.387 34.235	0.077 0.131 54.822	8.87 2.96 0.62	<0.001 0.007 0.58	0.936	84.4	161	<0.001

- high R^2 and F values and Sign.F <0.05: the prediction is indeed possible with a high degree of confidence
- predictors important and significant (t-value >1.5 and p-value<0.05)

Results: Accuracy of prediction

RQ	Model	MdAR	MAR
RQ1	Eff1	70.6	72
	FP1	160	195
	Eff2	117	159
	FP2	315	350
	Eff3	64.1	79.1
	FP3	171	159
RQ2	EffFP2	107	132
	EffFP3	53.2	69.7

Prior phase effort based models are better than **Function Points** based models

- the values of MdAR and MAR for Eff1 (Eff2, or Eff3) are more than two times lower wrt to FP1 (FP2 or FP3)

Results: Accuracy of prediction

RQ	Model	MdAR	MAR
RQ1	Eff1	70.6	72
	FP1	160	195
	Eff2	117	159
	FP2	315	350
	Eff3	64.1	79.1
	FP3	171	159
RQ2	EffFP2	107	132
	EffFP3	53.2	69.7

Combination of prior phase efforts and Function Points provided better results than using prior phase efforts or Function Points alone

Results: Research Questions

RQ	Comparison	Significant difference	Effect size
RQ1	Eff1 vs FP1	Yes	large
	Eff2 vs FP2	Yes	large
	Eff3 vs FP3	Yes	large
RQ2	EffFP2 vs Eff2	Yes	medium
	EffFP3 vs Eff3	No	small

Prior phase effort based models provided **statistically significant better** results than Function Points based models, with a large effect size

We can positively answer RQ1

Results: Research Questions (2)

RQ	Comparison	Significant difference	Effect size
RQ1	Eff1 vs FP1	Yes	large
	Eff2 vs FP2	Yes	large
	Eff3 vs FP3	Yes	large
RQ2	EffFP2 vs Eff2	Yes	medium
	EffFP3 vs Eff3	No	small

- the model combining early effort data, i.e., EFF_A , and the Function Points can provide **significantly better estimations** than the model based only on EFF_A (with a medium effect size)
- The difference is not statistical significant when combining EFF_A , EFF_D , and FP to predict $EFF_{D\&I}$

We can partially positively answer RQ2

Summary of results and contribution

- It is crucial to carefully collect information on effort spent during the project
- size measure remain important: combination of effort data and size measure can (significantly) improve the accuracy of effort estimations
 - size measure also employed for other purposes, e.g., productivity benchmarking

Threats to validity

Internal validity

- We did our best to control “extraneous factors”: use of the same questionnaires to collect info for all the applications and project managers were instructed on how to use the questionnaires

External validity

- Applications involved in study coming from one company
 - Replications with larger data sets from other companies

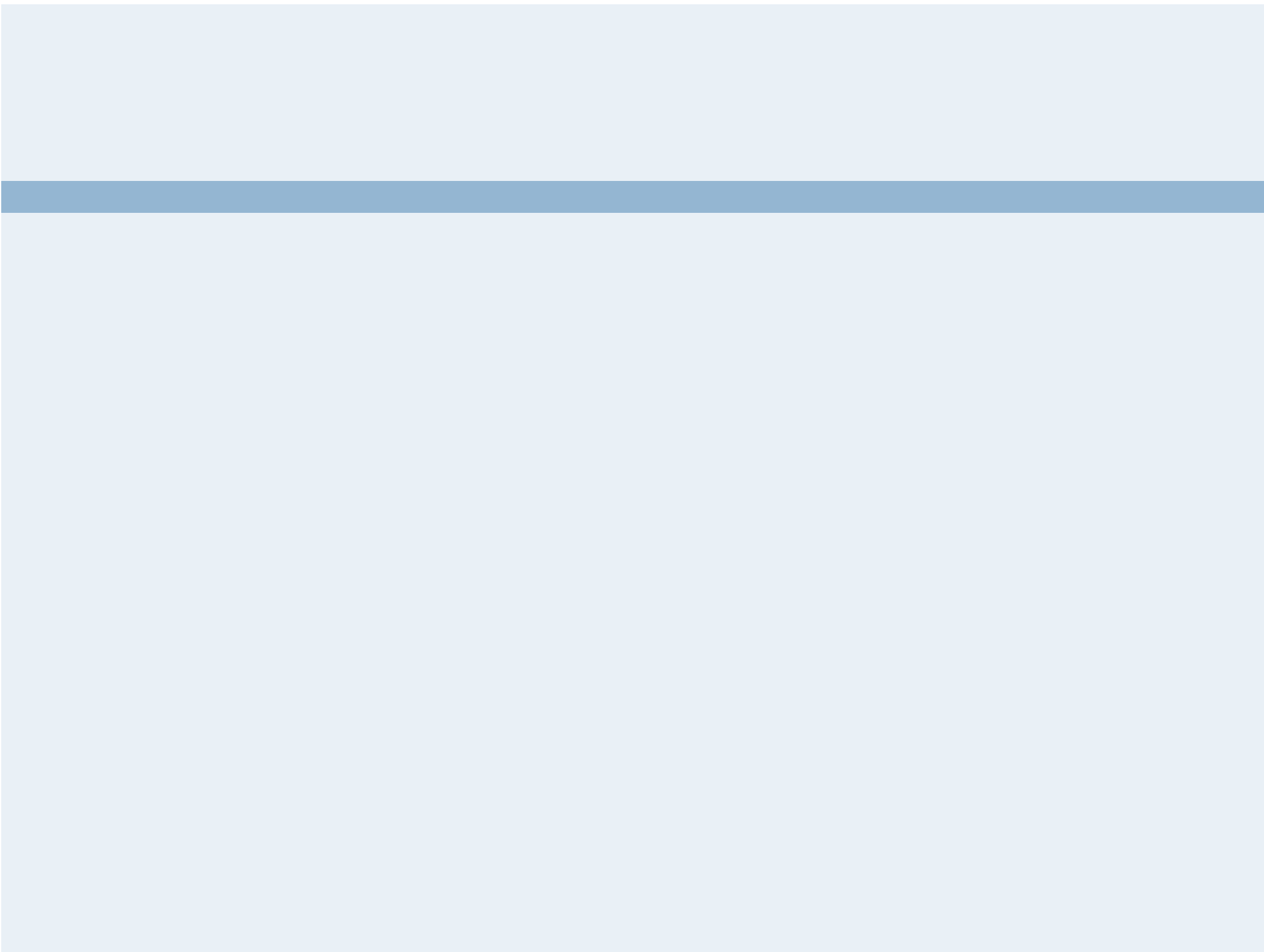
Threats to validity (2)

Construct validity

- Function Points Analysis has been applied for years by the company involved in our study
- We were very conscious about performing the collection of the information to determine the size measure and phase efforts (in a controlled and uniform way)

Conclusion validity

- Evaluation criteria allowed assessing in an objective way the effectiveness of the predictions
- We carefully applied the estimation methods and the statistical tests, verifying all the required assumptions



Differences with MacDonnel e Shepperd

- 16 projects collected from a single software company
- prior-phase effort data can be used to augment the estimation process already in place,
 - Our results confirm those of MacDonnel e Shepperd
- They did not use software size as an explanatory variable nor estimate total development effort
-

Differences with Tsunoda et al

- a model using software size, or early phase effort, or their combination
- Release 9 of ISBSG
- development phases: Planning, Requirement analysis, Planning and Requirement analysis
- Planning effort and Requirement analysis effort were not considered separately as predictor of total effort since they were characterized by multicollinearity
 - we exploit effort of prior-phases to predict the effort of the subsequent phases
 - we have predicted in some sense total effort (remaining phases)
 - Our results confirm and extend those of Tsunoda et al.